

Diagnostic Report

Report Title:	ROBOTIC ARM DELAY DIAGNOSTICS REPORT
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Ticket ID	#2437

Executive Summary

Purpose: The aim of report is to analyze the delays in surgical robotic arms particularly the rotate_joint command in terms of functional response times of the surgical robotic arms. The report spells out key essentials of the issue at hand, how the problems were verified, the deductions and the next steps.

Key Findings: The surgical robotic arms had all commands functional, but, there are delays in the working of the rotate_joint command. Rotate_joint was the slowest command at 0.15 s, close to the reported 0.18 s delay. move_arm and adjust_grip were within expected values. A simulated 20% improvement factor was applied; no code was changed.

Issue Description

Problem Statement: At Mercy General Hospital, a ticket was received from a customer. The issue was that delayed response times were observed by surgical robotic arms in their motion during surgery. These delays particularly occur in the rotate_joint command which causes noticeable delays during rotational motion of the arm for the correct positioning of the robotic gripper. This issue is worth considering as inefficiencies as such cause delays in surgical procedures.

Symptoms and Impact: The ticket flagged this issue as a "HIGH". Taking all information explicitly stated on the ticket and adding reasonable judgments made, the symptoms are no-

ticeable and observed delays when the robotic arm rotates to accurately position the gripper to enhance surgical procedures' success. An impact that could be very bad is inaccurate surgical procedures because of delayed and inaccurate positioning.

Client Information:

- **Hospital Name:** Mercy General Hospital
- **Location:** Rockville, Maryland.
- **Reported By:** Dr. Emily Chen, Senior Surgeon.

Diagnostic Process

Initial Observations: During my testing, I went on with a single idea and mindset; to suspend all judgments, presumptions and foreknowledge so that I will identify all problems not only the ones my mind had assumed. So, I tested the three commands. Upon testing, the move_arm command had a response time of 0.101 seconds, the rotate_joint command had a response time of 0.15 seconds and the adjust_grip command had a response time of 0.05 seconds.

Commands Tested: I tested the move_arm, rotate_joint and adjust_grip commands. During the testing, these command names were strings in a list. Then, a for loop was constructed such that for every command, the function that calculates response time is called and the response time is assigned to the respective command name.

Hypothesis:

- The rotate_joint command calculation might be redundant causing delays code execution and runtime.
- The rotate_joint command might have some problems with how algorithms are structured in a way that increases runtime and response time.

Tools and Techniques Used: I utilized Python, my main diagnostic tool. I mainly used the time module for outputting runtime and duration for code execution. All this happened in the Control System Diagnostic Notebook. I also used iterative testing for ensuring accurate results not just one- time confirmation.

Findings and Analysis

Response Time Data:

Command	Expected response Time	Re- sponse Time	Initial Time	Response	Optimized response Time	Re- sponse Time
move_arm	0.10 seconds		0.101 seconds		0.08 seconds	
rotate_joint	0.18 seconds		0.15 seconds		0.12 seconds	
adjust_grip	0.09 seconds		0.05 seconds		0.04 seconds	

Analysis of Findings:

- The rotate_joint command had a response time of 0.15 seconds which is closer to the flagged, delay duration of 0.18 seconds. This shows that the response time has to still be considered since, it is not far from the delay value.
- The adjust_grip and move_arm commands work well but, since no problems or issues were witnessed with those commands, I assume they are functioning as expected with their response times of 0.05 and 0.101 seconds respectively compared to the expected values 0.09 and 0.10 seconds respectively.

Optimizations and Solutions

Code Modifications: There were no code changes to affect real time response time of the robotic commands. However, there was a function that optimized the response time obtained by multiplying it with a conversion factor for all the commands. That was an arithmetic optimization; it showed how the values would change if the effect of changing the code or fixing the bug will be equal to multiplying by a factor.

Impact of Optimizations: I also think that we should test the commands independent of each other. We should not only test the issue by running tests to obtain runtime but, also, inspecting the code's algorithmic makeup so that we see what the root cause is. Also, we have to investigate the method employed in the calculation to validate its effectiveness.

Conclusion

Summary of Findings: The issue is a very serious problem as it risks surgical operation accuracy. Also, its solution will only be fully fledged if we suspend all judgments and work to investigate the root cause of the problem. rotate_joint is the most likely source of the delay and the priority for investigation, but the root cause hasn't been found yet

Overall Impact: When the issue is resolved, surgical operations will take place with accuracy. Surgical robots will accurately position grips for surgeries. Delays in surgeries and more human interaction will be curbed. If the root cause is found, delays will be reduced.

Next Steps:

- Investigate the root cause of the root cause of the problem.
- Take repeated measurements and averages.
- I will inspect rotate_joint code for redundant calculations.